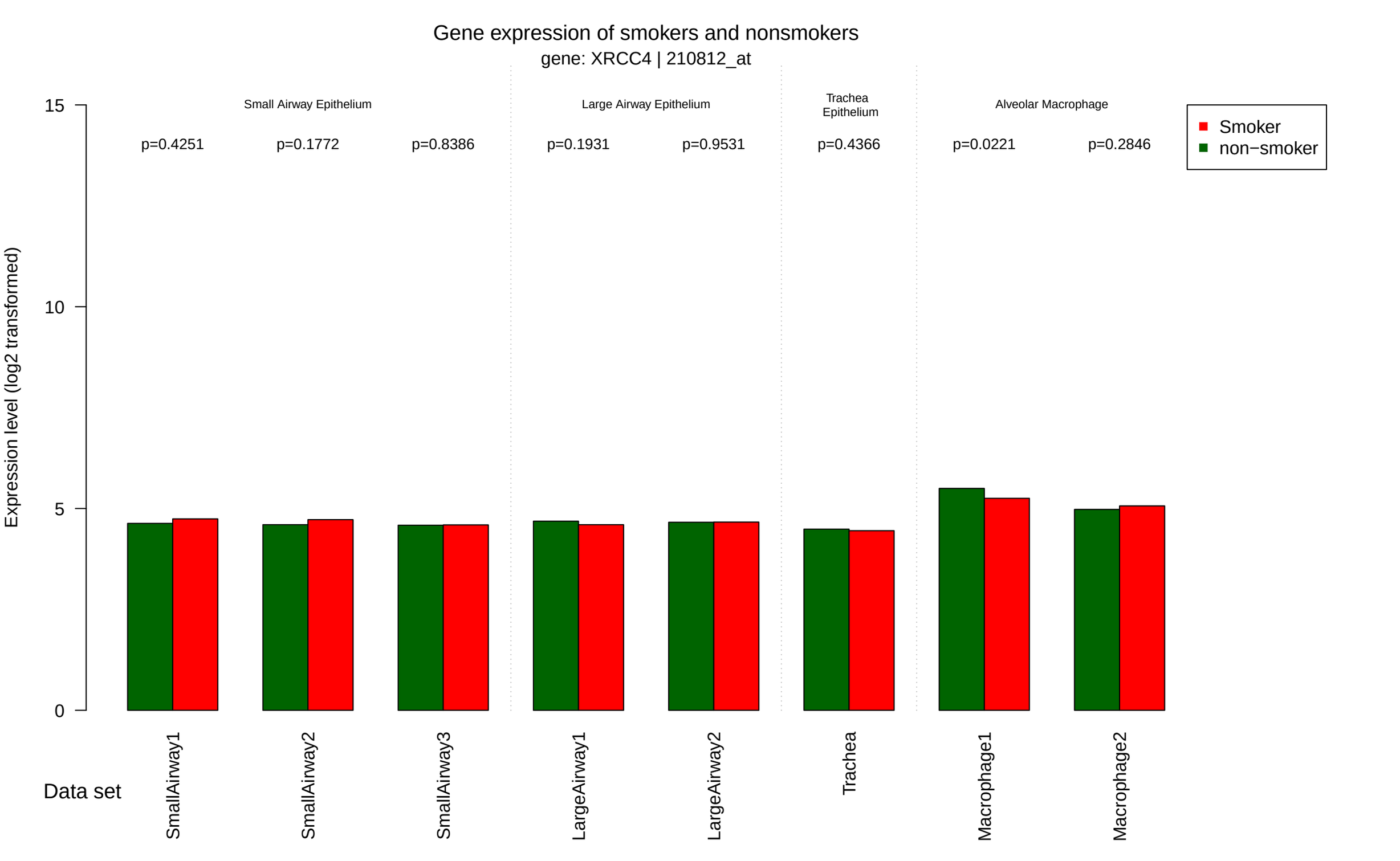
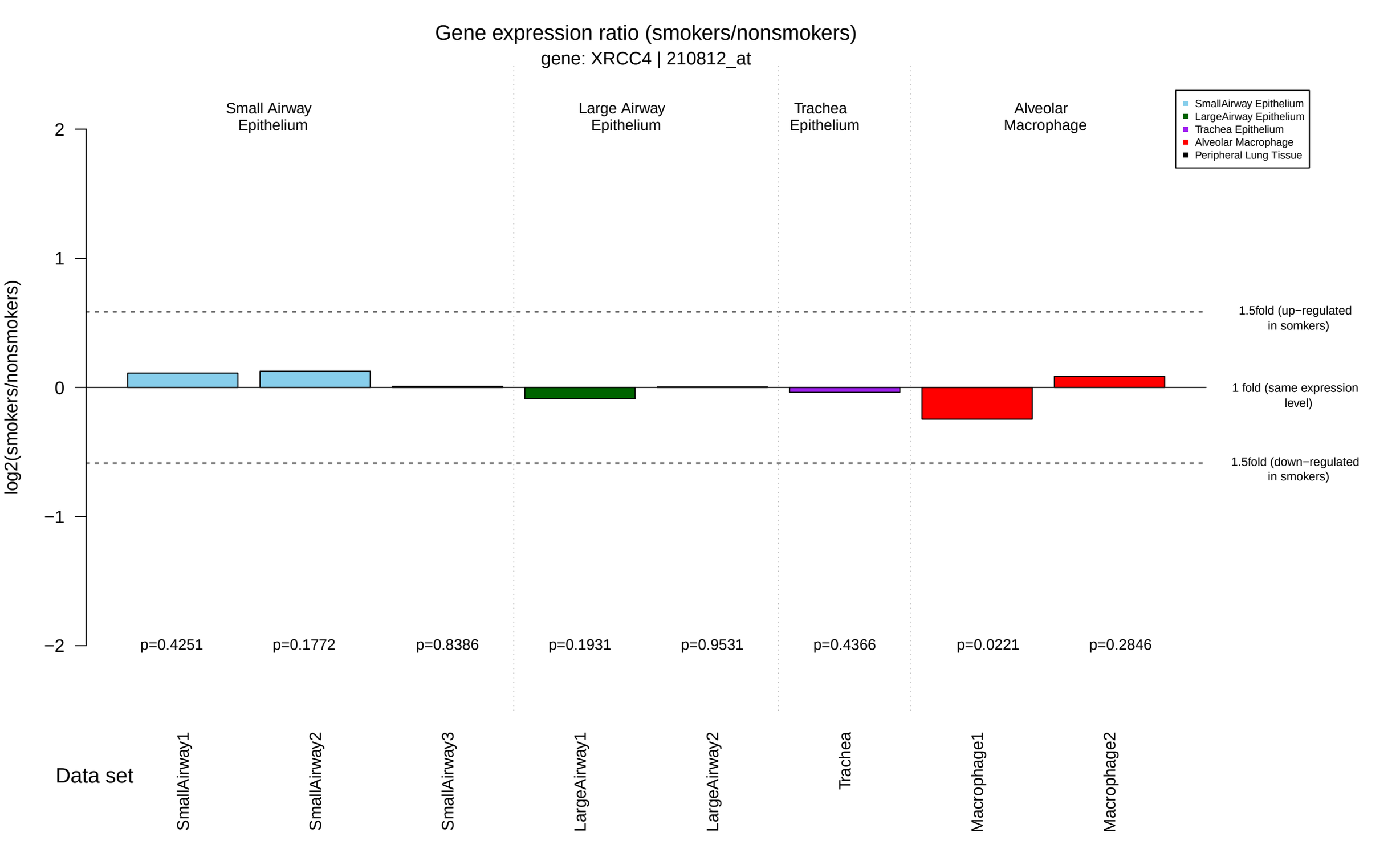
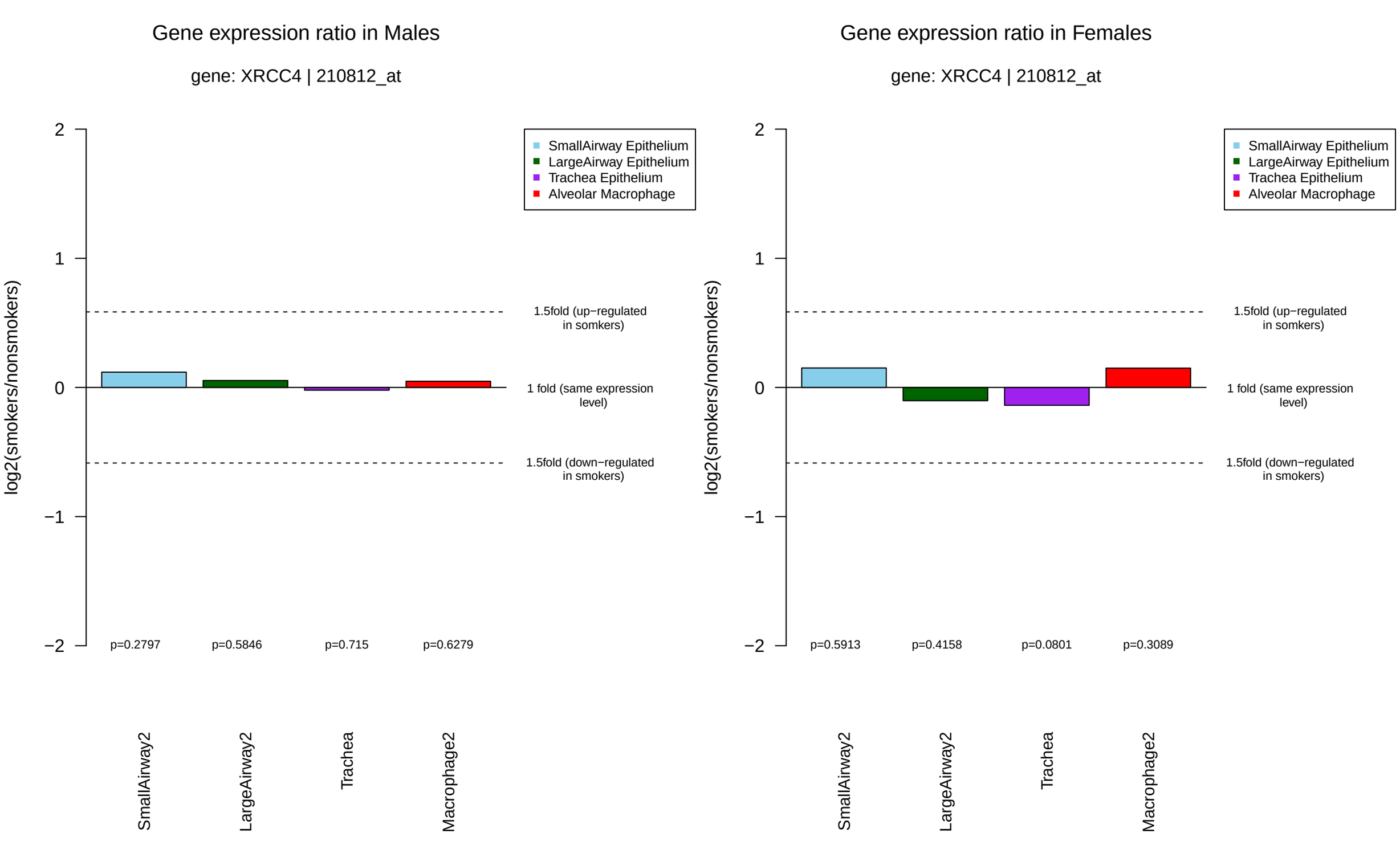
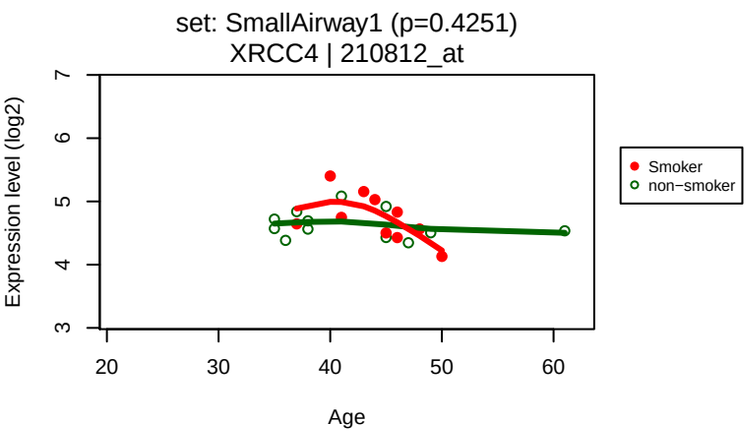
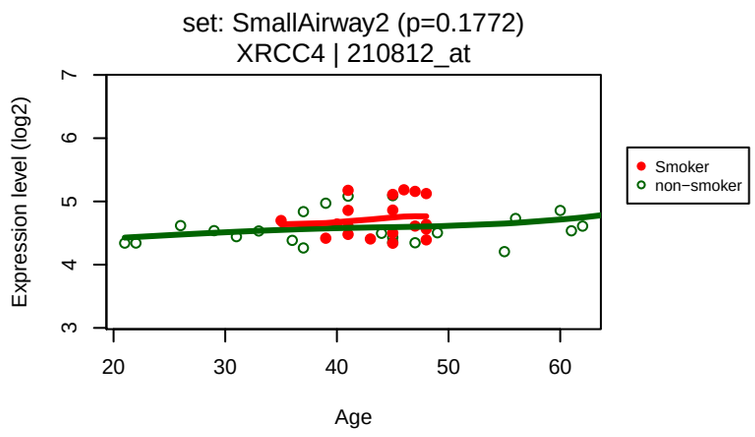
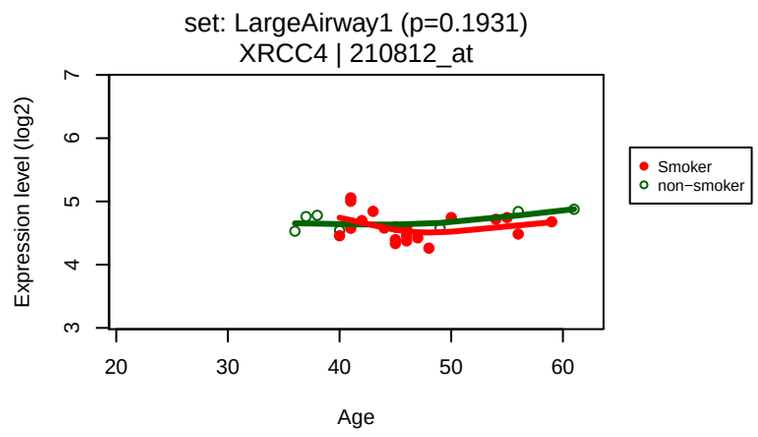

Gene expression of smokers and nonsmokers
gene: XRCC4 | 210812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4251
p=0.1772
p=0.8386
p=0.1931
p=0.9531
p=0.4366
p=0.0221
p=0.2846
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

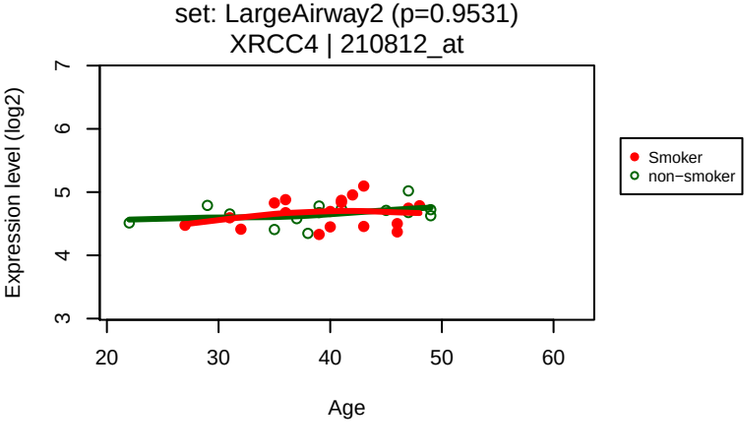

Gene expression ratio (smokers/nonsmokers)
gene: XRCC4 | 210812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4251
p=0.1772
p=0.8386
p=0.1931
p=0.9531
p=0.4366
p=0.0221
p=0.2846
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

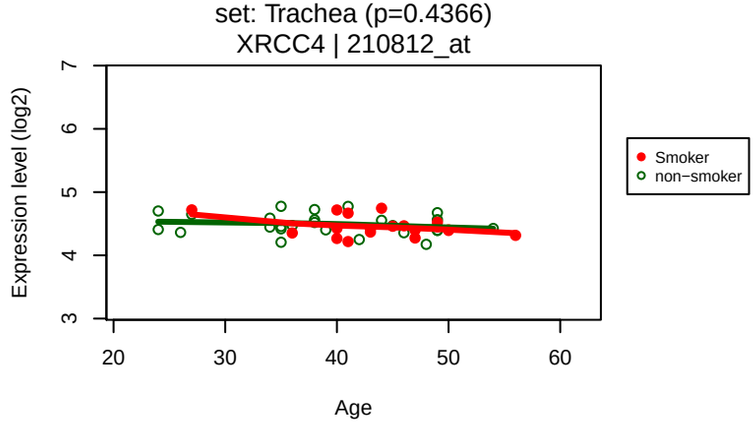

Gene expression ratio in Males
gene: XRCC4 | 210812_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2797
p=0.5846
p=0.715
p=0.6279
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XRCC4 | 210812_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5913
p=0.4158
p=0.0801
p=0.3089
SmallAirway2
LargeAirway2
Trachea
Macrophage2

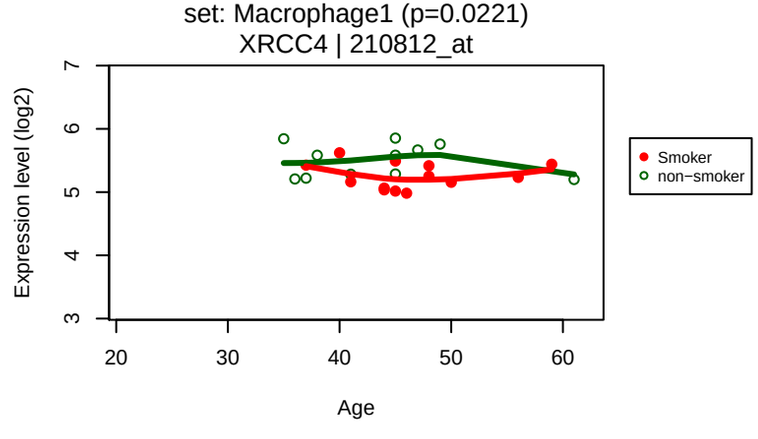

set: SmallAirway1 (p=0.4251)
XRCC4 | 210812_at
Expression level (log2)
Age
Smoker
non−smoker

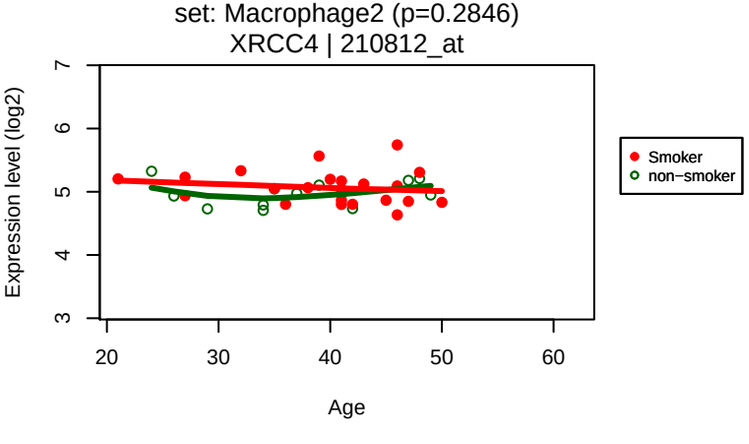

set: SmallAirway2 (p=0.1772)
XRCC4 | 210812_at
Expression level (log2)
Age
Smoker
non−smoker

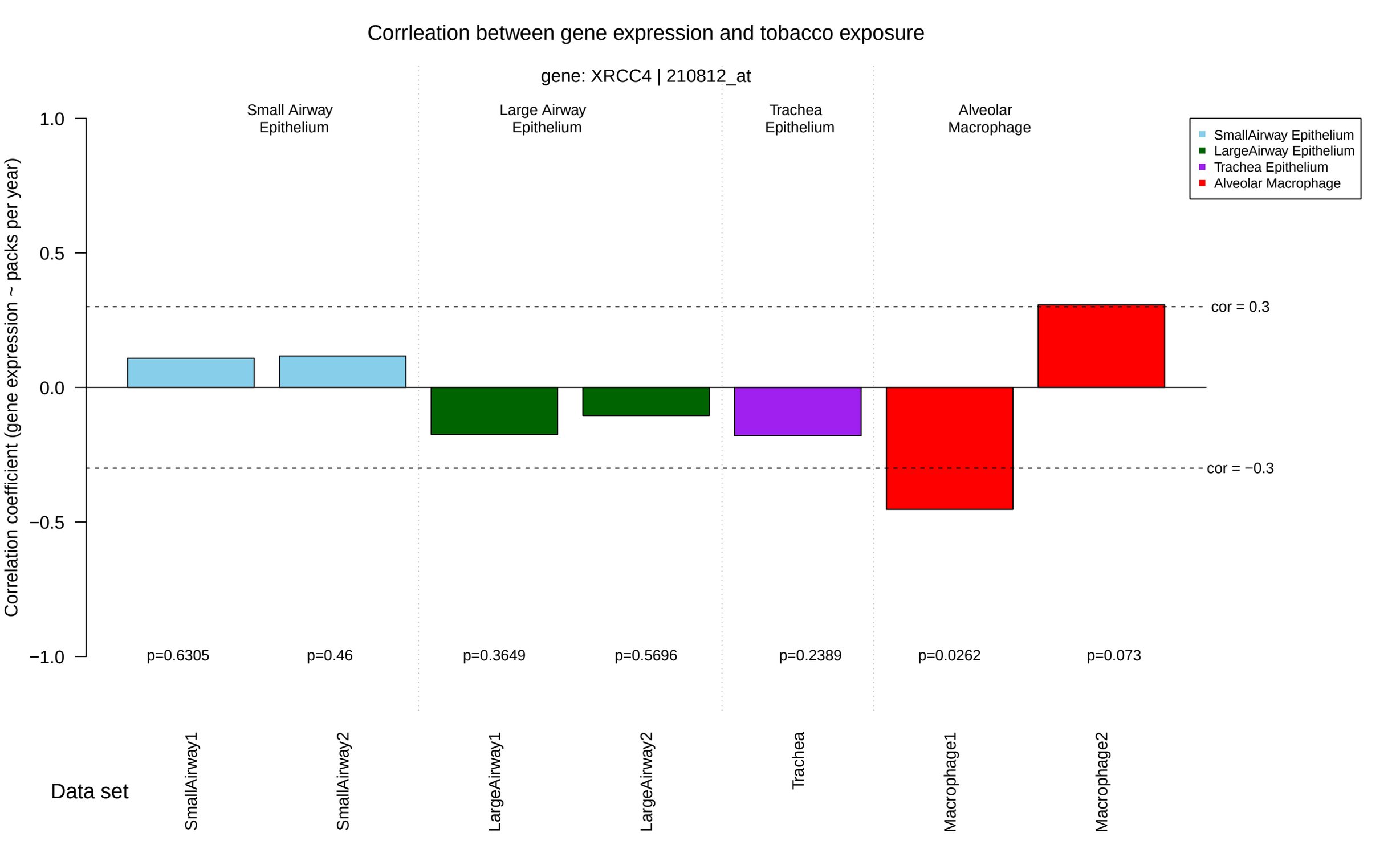

set: LargeAirway1 (p=0.1931)
XRCC4 | 210812_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9531)
XRCC4 | 210812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4366)
XRCC4 | 210812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0221)
XRCC4 | 210812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2846)
XRCC4 | 210812_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XRCC4 | 210812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6305
p=0.46
p=0.3649
p=0.5696
p=0.2389
p=0.0262
p=0.073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2